



Artificial Intelligence Engineering
Department



تحکم الکی =
دکاء بالک

مقدمه

LECTURE - (1)

INTRODUCTION TO AUTOMATIC CONTROL

10/19/2025

Asst.Prof Fouad SALHA

Automatic Control

Academic year 2025/2026

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introduction to automatic control Outline

Outline

- What is automatic control ? System
- Application exemples أمثلة التطبيقات
- Control systems technologies تقنيات التحكم بالأنظمة
- Open-loop vs. closed-loop control اللووب المفتوح ضد المغلق
- Control systems design requirement متطلبات تصميم الأنظمة التحكم
- Course information معلومات الكورس

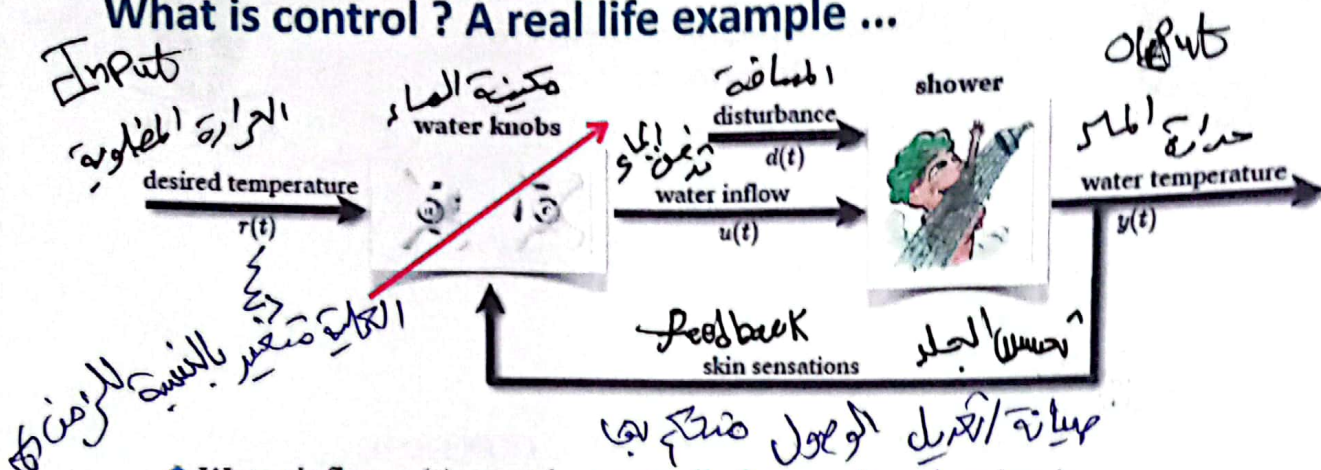
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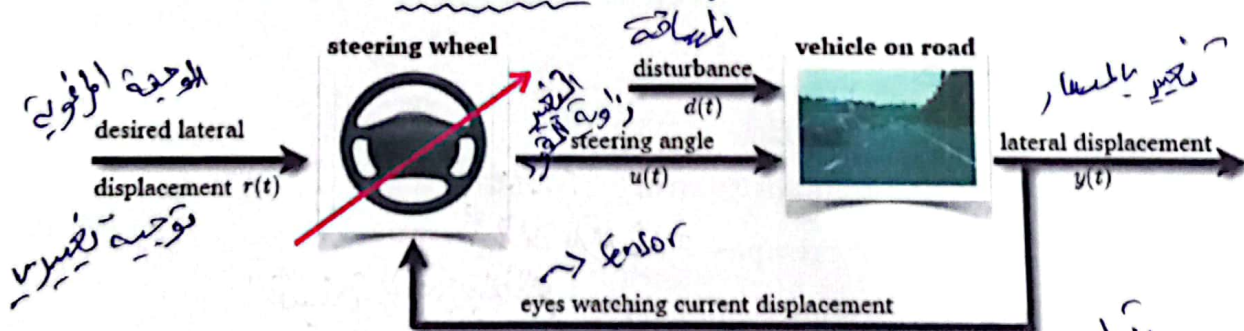
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What is control ? A real life example ...



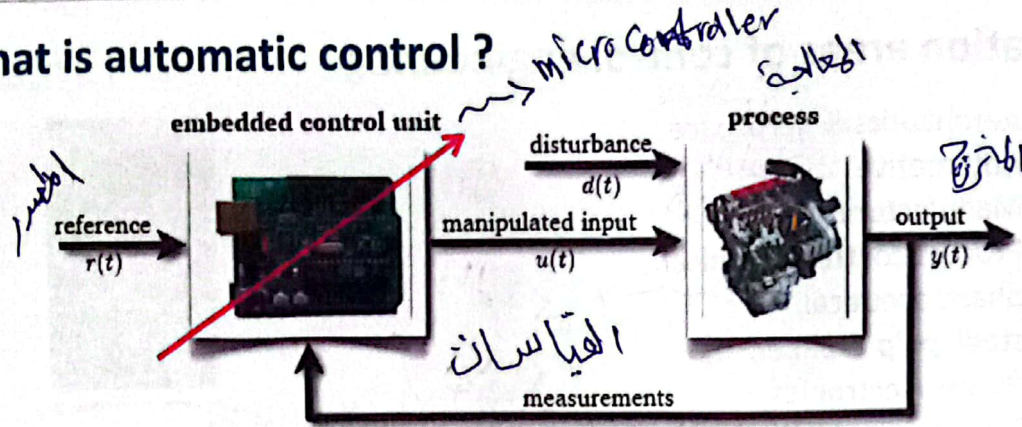
- Water inflow $u(t)$ must be **controlled** to reach and maintain the desired temperature $r(t)$
- Sensors on skin **measure** water temperature $y(t)$
- Water inflow $u(t)$ manipulated so that $y(t) \approx r(t) \dots$
- ... in spite of flow and temperature fluctuations $d(t)$

What is control ? A real life example ...



- Steering wheel must be **controlled** to reach and maintain the desired lateral displacement $r(t)$ within the lane (e.g.: staying in the middle of the lane)
- Eyes **measure** current lateral displacement $y(t)$
- Steering wheel $u(t)$ manipulated so that $y(t) \approx r(t) \dots$
- ... in spite of changes of road curvature and of $r(t)$

What is automatic control ?



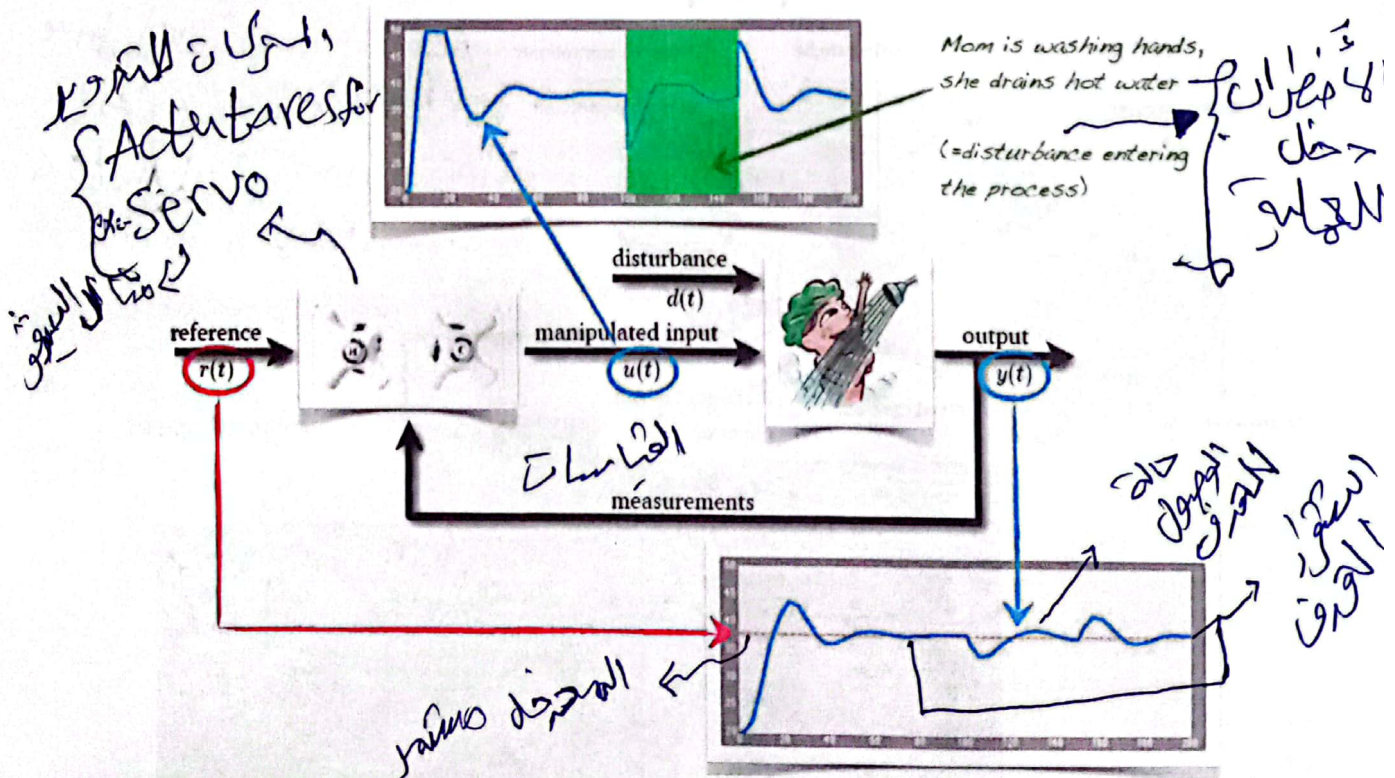
How to **control** the inputs $u(t)$ to the process **automatically** to make the output $y(t)$ **track** the given reference $r(t)$?

performance

How to exploit the **measurements** of $y(t)$ to track the reference $r(t)$ in spite of **disturbances** $d(t)$ acting on the process ?

robustness

What is automatic control ?

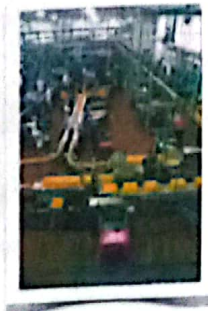


Application areas of control engineering

- Aeronautics & aerospace
- Automotive
- Manufacturing
- Process control (chemical, pharmaceutical, steel, pulp & paper, ...)
- Power electronics
- Telecommunications
- Environmental systems
- Financial engineering
- Supply chains
- Power networks

مراقبة
العمليات
Interf.

Cruise
Control



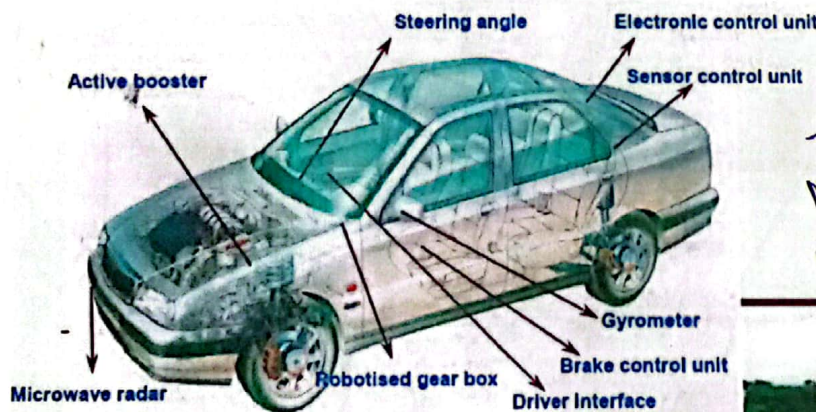
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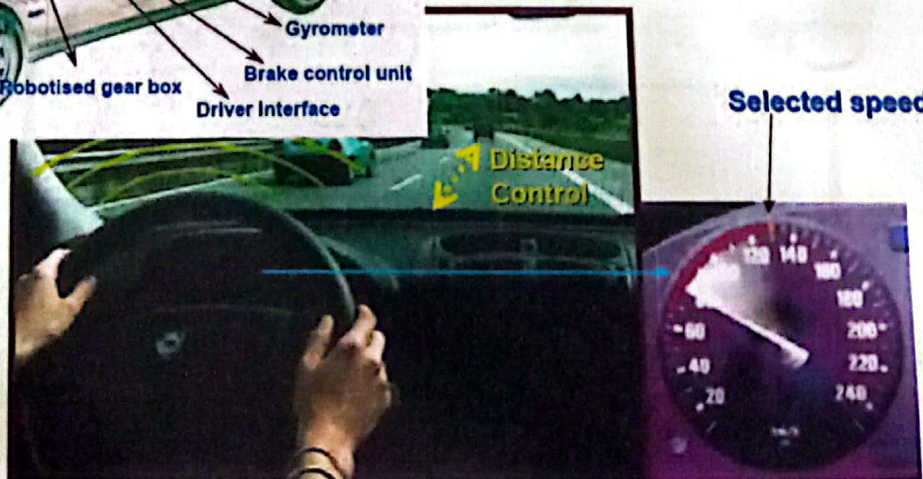
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Adaptive Cruise Control (ACC)



(ACC) is an advanced drive assistance system (ADAS), that automatically adjusts a vehicle's speed to reach maintain.



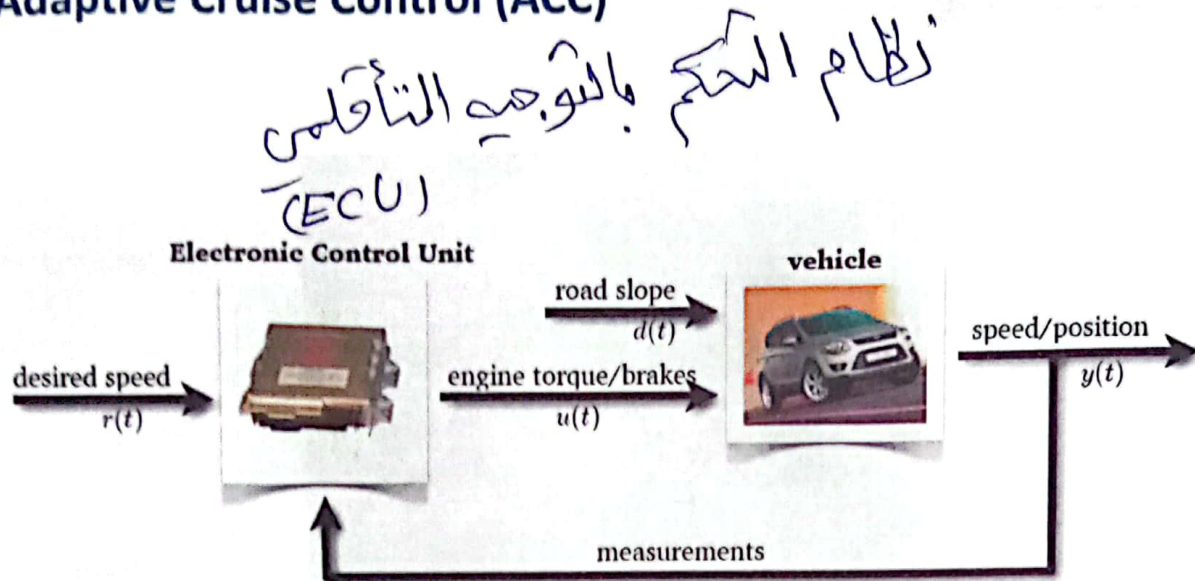
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Adaptive Cruise Control (ACC)



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Lane Warning

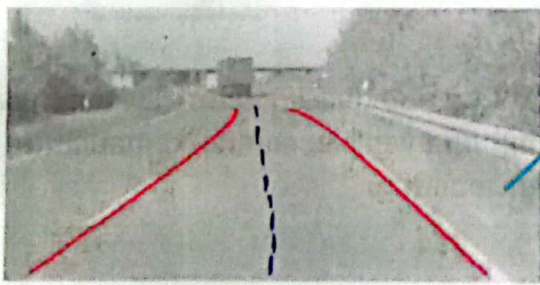
Optical sensor detects and tracks lane markings

- Incipient lane change signalled to the driver by:

- Haptic signal on driver seat (simulation of rumble strip)
- Haptic feedback on steering wheel
- Acoustic directional signals

Sound or sign on screen

Lane can be automatically maintained



Haptic Feedback



electric power steering

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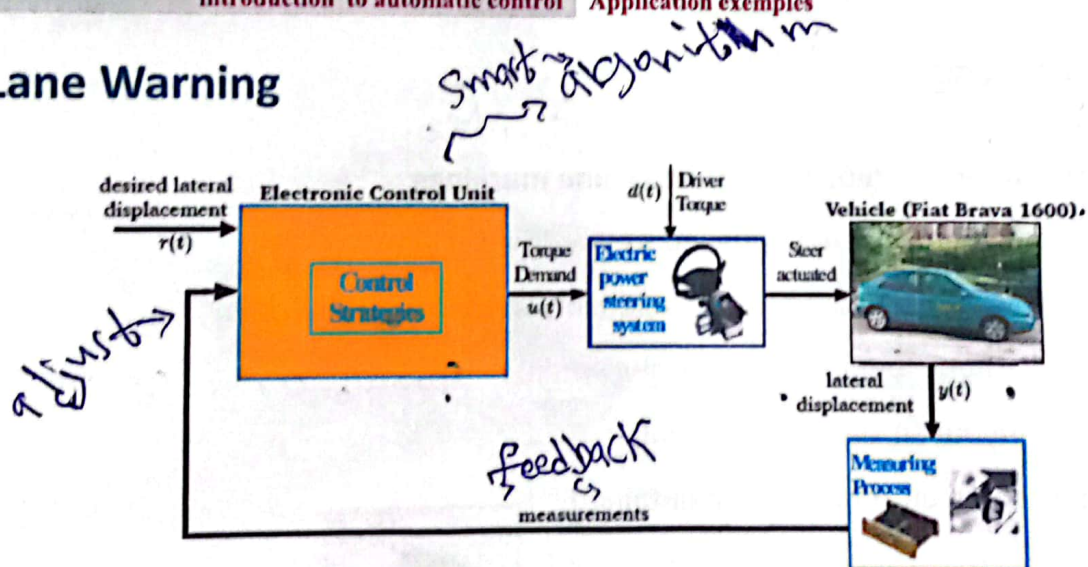
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Lane Warning

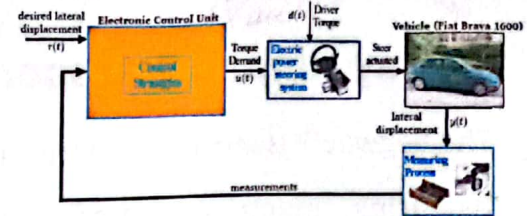


Lane Warning



- This diagram illustrates a closed-loop feedback control system used for automatic lateral guidance (steering) of a vehicle, aiming to match the actual vehicle position to a desired reference path.

Lane Warning



Component	Role in the System	Key Variables
Electronic Control Unit (ECU)	The Controller or "Brain." It calculates the necessary corrective action (Torque Demand $u(t)$) by comparing the desired path $r(t)$ with the measured position $y(t)$.	Input: Desired Displacement $r(t)$, Feedback Output: Torque Demand $u(t)$
Electric Power Steering (EPS)	The Actuator . It converts the electrical command $u(t)$ into mechanical steering action (Steer actuated). It must also account for or reject the Driver Torque $d(t)$ (a disturbance).	Input: $u(t)$ and Driver Torque $d(t)$
Vehicle (Fiat Brava 1600)	The Plant or Process being controlled. Its dynamics determine how the steering input affects the lateral position.	Output: Actual Lateral Displacement $y(t)$
Measuring Process	The Sensor/Feedback system. It measures the physical lateral displacement $y(t)$ and converts it into an electrical signal (measurements) for the ECU.	Output: Measurements (Feedback signal)



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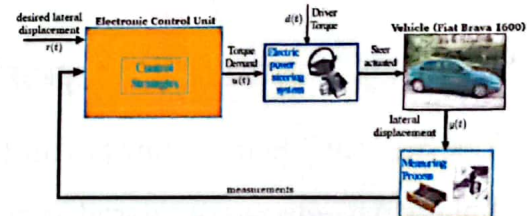
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Lane Warning

Feedback Loop Operation

The system functions by continuously minimizing the error between the desired state and the actual state:

- **Reference ($r(t)$):** The target path is set (e.g., stay in the center of the lane).
- **Feedback ($y(t)$):** The vehicle's actual position is measured.
- **Control Strategies:** The ECU generates a command $u(t)$ proportional to the difference ($r(t)-y(t)$).
- **Correction:** $u(t)$ commands the EPS to steer the vehicle, correcting the lateral displacement $y(t)$ until the error approaches zero.
- This configuration is typical for advanced driver assistance systems (ADAS) like Lane-Keeping Assist (LKA) or full autonomous steering.



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introduction to automatic control Application exemples

الموازن
الإنسان
ناقل

Segway Human Transporter

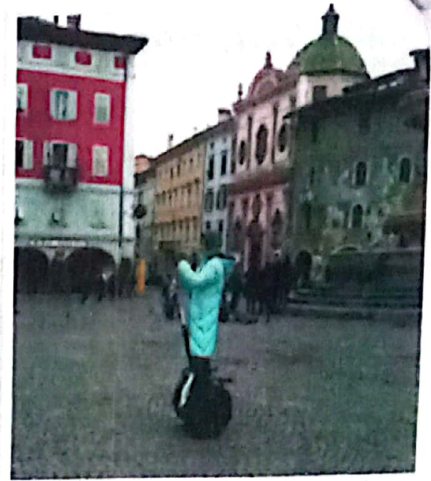
الموازن
الإنسان

The Segway™ Human Transporter (HT) is the first self-balancing, electric-powered transportation device. With dimensions no larger than the average adult body and the ability to emulate human balance.



Gyroscopes and tilt sensors in Segway HT monitor a user's center of gravity at about 100 times a second. When a person leans slightly forward, Segway HT moves forward. When leaning back, Segway HT moves back.

مبدأ التوازن البشري



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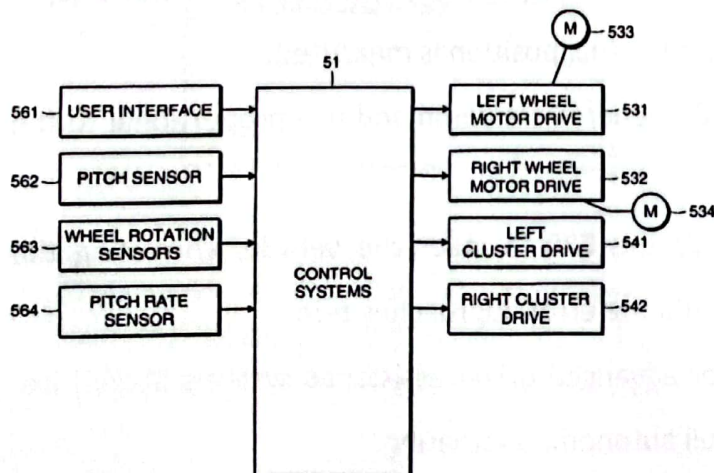
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Segway Human Transporter

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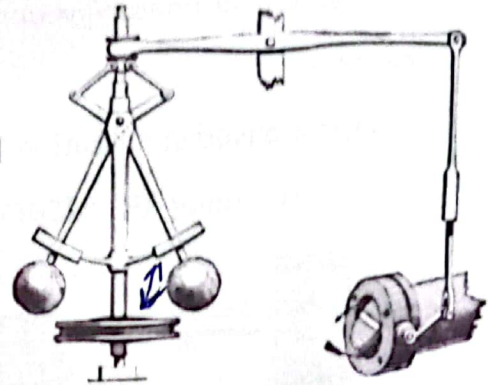
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Short historical notes

- **Water level controllers date back to ancient Greece**
(Ktesibios of Alexandria, water clock, 270 B.C.), (Philon of Byzantium, 250 B.C.) (Heron of Alexandria, ≈ 100 A.D.)
- **Other examples in the 17th-18th century**
(Cornelis Drebbel's incubator, 1620), (Edmund Lee's self-regulating wind machine, 1745) (Thomas Mead's lift tender and speed controller, 1787)
- **James Watt's flyball governor (1788)**
Used to regulate the speed of steam engines.
If speed increases, flyballs spread apart and the steam flow through the throttle is reduced
And vice versa.
- **Frequency domain (1930-1950)**
(H. Nyquist, H.W. Bode, N.B. Nichols)
- **State-space and optimal control (1950-1980)**
(R. Kalman, R. Bellman)
- **Nonlinear, robust, adaptive, optimization-based, (...) control (1980-today)**



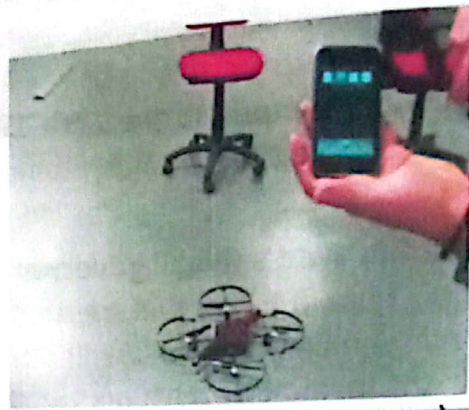
Aeronautics and aerospace

- **Control systems heavily used in:**
 - Aircrafts (roll, pitch, yaw)
 - Helicopters
 - Satellites
 -

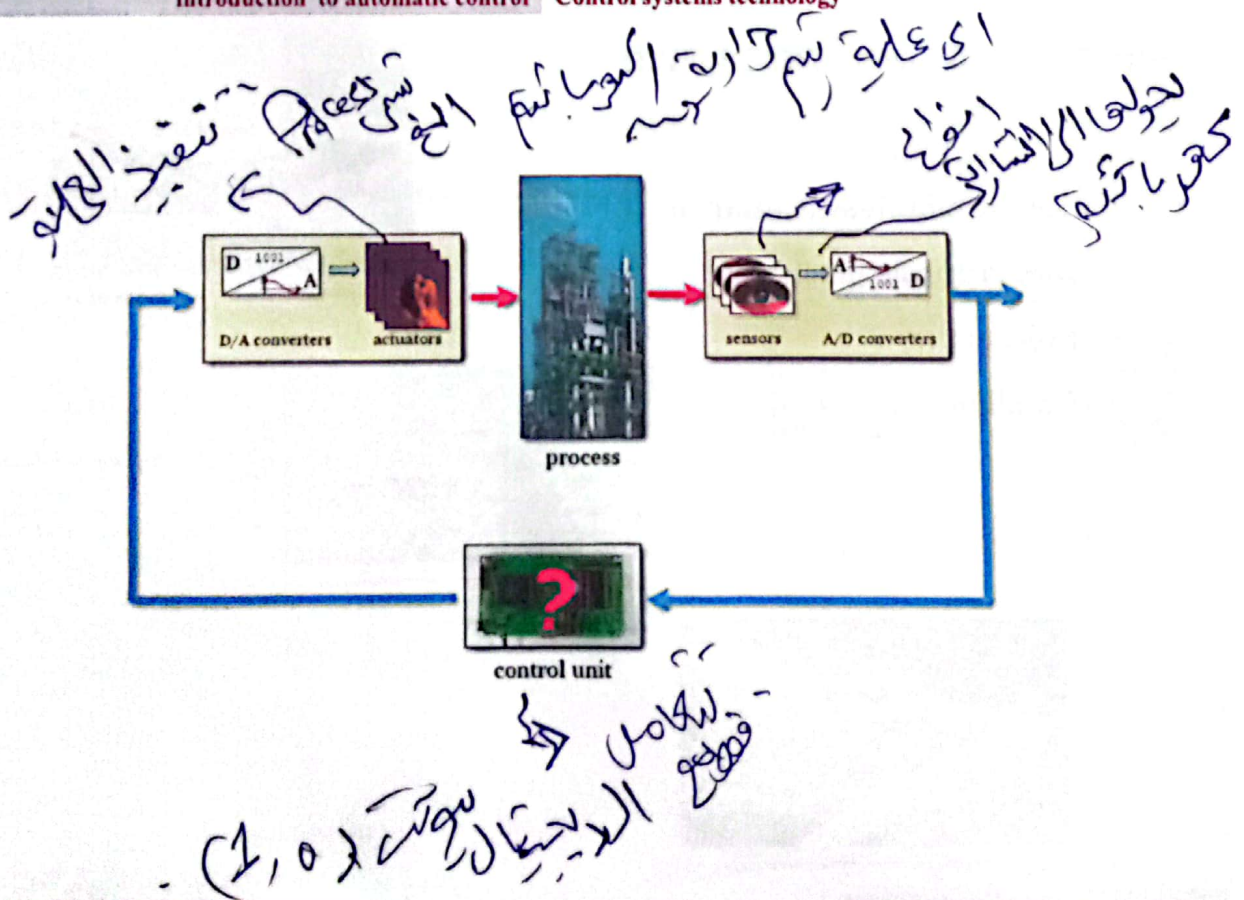
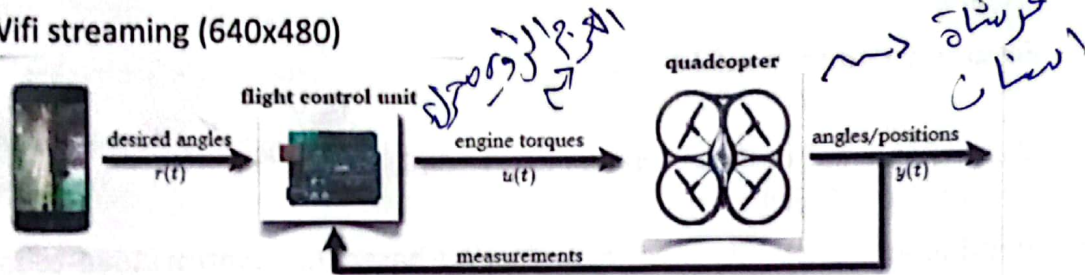


Aeronautics and aerospace. Quadcopter example

- Parrot AR Drone
- 6 DoF Inertial Measurement Unit (3 accelerometers, 3 gyros)
- 468MHz ARM9 CPU running Linux
- Ultrasound altimeter/ground detector Two cameras:



- Vision-based stabilization (176x144)
- Wifi streaming (640x480)



Most used sensors and actuators in control systems

Sensors



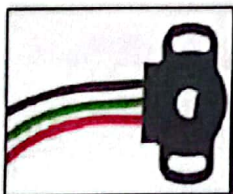
- ❖ temperature
- ❖ pressure
- ❖ flow
- ❖ level
- ❖ velocity, position
- ❖ acceleration
- ❖ force (strain) / deformation

Actuators

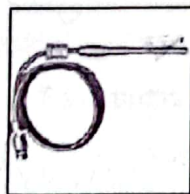


- ❖ electrical motors (DC, brushless, step)
- ❖ pumps
- ❖ valves
- ❖ heaters

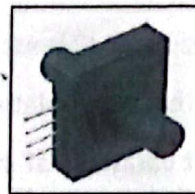
Sensors and actuators in control systems



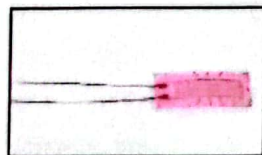
angular position



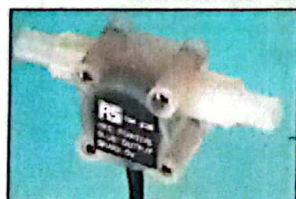
thermocouple



pressure sensor



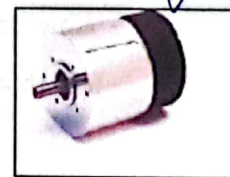
strain gage



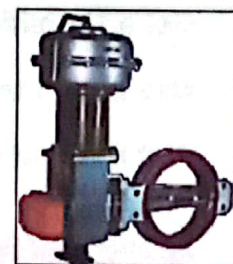
liquid flow sensor



pump



brushless motor

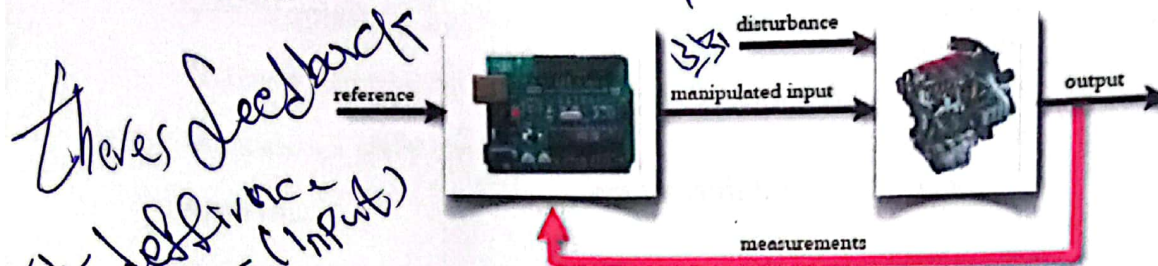


valve

Open-loop vs. closed-loop control

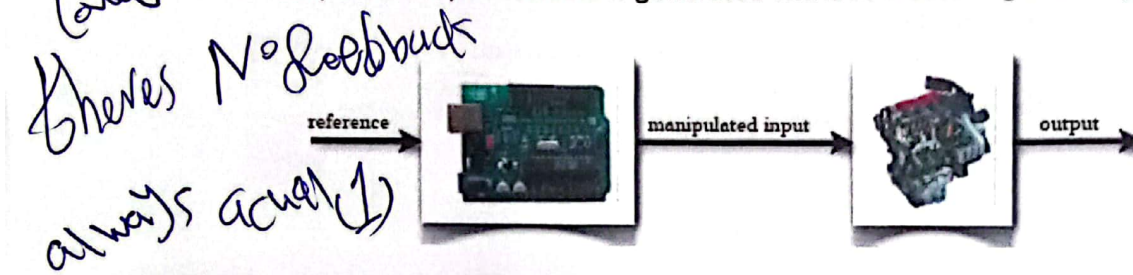
- Closed-loop control (**feedback control**):

Measurements of the output variables are fed back to the process through the controller



Open-loop control (**feedforward control**)

The manipulated input variable is generated without measuring the output variable



How to design a (modern) control system

- Understand the automation problem: *ادخال*

- Which variables can be manipulated by actuators ?
- What are the output variables of interest ?
- What should we measure ? *متغيرات المدخلات*
- Which are the disturbances ? *متغيرات المداخلات*

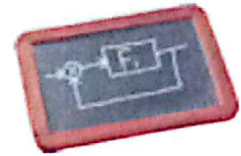
- Get a reliable **simulation model**
- Get a simplified **mathematical model** of the main process dynamics
- Use design techniques to synthesize the **control algorithm** *فصل ٢٠*
- Test in simulation, **validate** on real process *Testing it*

Control design requires a dynamical model of the open-loop process

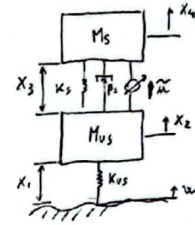
Pass

What you will learn in this course

- The tools to study the mathematical properties of dynamical systems (control theory is often called "systems theory").



- Building simple dynamical models of common physical processes.



- Computer-aided tools for analysis, simulation, and control of dynamical systems (in MATLAB™)



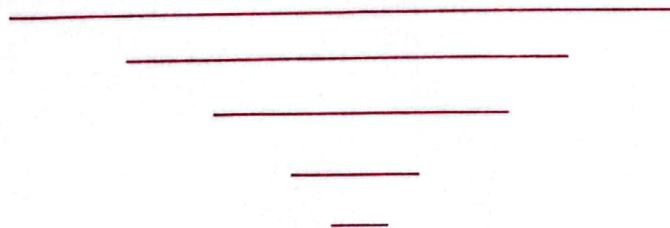
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What you will learn in this course

- Analysis of dynamical systems in continuous time (differential equations, Laplace transform, stability)
- Models of dynamical systems (electrical, mechanical, hydraulic,...)
- State-feedback control synthesis in the time domain (pole-placement)



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End of Lectures 1



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